1.AIM: Create an XML document for a Library Management System with the following elements:

library

- **book**
  - **title**
  - **author**
  - **price**

a. **Ensure the XML is well-formed.**

b. **Intentionally introduce one error (missing closing tag) and observe the browser error.**

c. **Correct the error and re-validate.**

CODE:#XML code

```
<library>
<book>
<title>The silent patient</title>
<author>Alex Michaelides</author>
<price>254</price>
</book>
<book>
<title>The Palace of Illusions</title>
<author>Chitra Banerjee Divakaruni</author>
<price>325</price>
</book>
<book>
<title>White Nights</title>
<author>Fyodor Dostoyevsky</author>
<price>111</price>
</book> </library>
```

a. this xml file is validated successfully and there is no error.

b.I intentionally missed the semi colon at end tag of last book price and tried to validate it,the validation is failed and the following error is displayed

```
Unable to perform operation.
Input is not valid XML. Error on line: 16
Error: The element type "price" must be terminated by the matching end-tag "</price>".
```

c.The code is corrected and revalidated again successfully.

2.AIM: Design an XML Schema (XSD) for the Library XML created in Task 1.

CODE: #XSD code

```
<?xml version="1.0" encoding="UTF-8"?>

<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema">

<xs:element name="library">

  <xs:complexType>

  <xs:sequence>

    <xs:element name="book" minOccurs="0" maxOccurs="unbounded">

      <xs:complexType>

      <xs:sequence>

        <xs:element name="title" type="xs:string"/>

        <xs:element name="author" type="xs:string"/>

        <xs:element name="price" type="xs:positiveInteger"/>

      </xs:sequence>

      </xs:complexType>

    </xs:element>

  </xs:sequence>

  </xs:complexType>

</xs:element>

</xs:schema>
```

3.AIM: Write DTD and XSD for the following XML code

Bookstore.xml

```
<bookstore>
  <book price="730.54" ISBN="string" publicationdate="2016-02-27">
  <title>string</title>
    <author>
      <first-name>string</first-name>
      <last-name>string</last-name>
    </author>
    <genre>string</genre>
  </book>
  <book price="6738.774" ISBN="string">
    <title>string</title>
    <author>
      <first-name>string</first-name>
      <last-name>string</last-name>
    </author>
  </book>
</bookstore>
```

CODE: #DTD code

```
<!ELEMENT bookstore (book*)>
<!ELEMENT book (title, author, genre?)>
<!ELEMENT title (#PCDATA)>
<!ELEMENT author (first-name, last-name)>
<!ELEMENT first-name (#PCDATA)>
<!ELEMENT last-name (#PCDATA)>
<!ELEMENT genre (#PCDATA)>
```

```
<!ATTLIST book     price CDATA
#REQUIRED
   ISBN CDATA #REQUIRED
   publicationdate CDATA #IMPLIED
>
```

#XSD code

```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:element name="bookstore">
   <xs:complexType>
   <xs:sequence>
      <xs:element name="book" minOccurs="0" maxOccurs="unbounded">
         <xs:complexType>
         <xs:sequence>
            <xs:element name="title" type="xs:string"/>
            <xs:element name="author">
               <xs:complexType>
               <xs:sequence>
                  <xs:element name="first-name" type="xs:string"/>
                  <xs:element name="last-name" type="xs:string"/>
               </xs:sequence>
               </xs:complexType>
            </xs:element>
            <xs:element name="genre" type="xs:string" minOccurs="0"/>
         </xs:sequence>
```

```
        <xs:attribute name="price" type="xs:float"/>
        <xs:attribute name="ISBN" type="xs:string"/>
        <xs:attribute name="publicationdate" type="xs:date" use="optional"/>
        </xs:complexType>
      </xs:element>
    </xs:sequence>
    </xs:complexType>
  </xs:element>
</xs:schema>
```

4.AIM: Modify the Task-3 XSD by adding attributes:

book element must have:

i. ISBN (required)
ii. year (optional)
iii. Do:

Update the XML document.
Modify the XSD to support attributes.
Test validation by removing the required attribute and observe errors.

CODE: The modified XML file is

```
<bookstore>
  <book price="730.54" ISBN="string" year="2016" publicationdate="2016-02-27">
  <title>string</title>
    <author>
      <first-name>string</first-name>
      <last-name>string</last-name>
    </author>
    <genre>string</genre>
  </book>
  <book price="6738.774" ISBN="string" year="2018" >
```

```
      <title>string</title>
      <author>
        <first-name>string</first-name>
        <last-name>string</last-name>
      </author>
    </book>
  </bookstore>
```

The modified XSD file is

```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:element name="bookstore">
  <xs:complexType>
    <xs:sequence>
      <xs:element name="book" minOccurs="0" maxOccurs="unbounded">
        <xs:complexType>
          <xs:sequence>
            <xs:element name="title" type="xs:string"/>
            <xs:element name="author">
              <xs:complexType>
                <xs:sequence>
                  <xs:element name="first-name" type="xs:string"/>
                  <xs:element name="last-name" type="xs:string"/>
                </xs:sequence>
              </xs:complexType>
            </xs:element>
            <xs:element name="genre" type="xs:string" minOccurs="0"/>
          </xs:sequence>
          <xs:attribute name="price" type="xs:float"/>
```

```
            <xs:attribute name="ISBN" type="xs:string" use="required"/>
            <xs:attribute name="year" type="xs:gYear" use="optional"/>
            <xs:attribute name="publicationdate" type="xs:date" use="optional"/>
          </xs:complexType>
        </xs:element>
      </xs:sequence>
    </xs:complexType>
  </xs:element>
</xs:schema>
```

By validating the modified files after removing the required attributes in XSD file we will get the following error

- Cvc-complex-type.4: Attribute 'ISBN' Must Appear On Element 'book'., Line '2', Column '64'.
- Cvc-complex-type.4: Attribute 'ISBN' Must Appear On Element 'book'., Line '10', Column '38'.

5.AIM: Add restrictions to Task-4 XSD for the following:

b. **year must be between 2000 and 2025**
c. **price must be greater than 100 Do:**

Apply xs:restriction in XSD.
Test with both valid and invalid values.
Record validation errors.

CODE:After adding restrictions to the attributes the XSD code is

```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:element name="bookstore">
  <xs:complexType>
    <xs:sequence>
      <xs:element name="book" minOccurs="0" maxOccurs="unbounded">
        <xs:complexType>
```

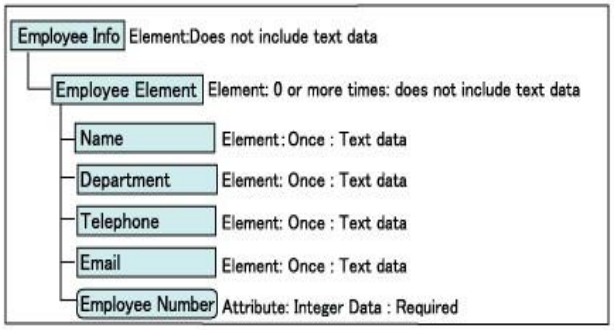
```
    <xs:sequence>
      <xs:element name="title" type="xs:string"/>
      <xs:element name="author">
        <xs:complexType>
          <xs:sequence>
            <xs:element name="first-name" type="xs:string"/>
            <xs:element name="last-name" type="xs:string"/>
          </xs:sequence>
        </xs:complexType>
      </xs:element>
      <xs:element name="genre" type="xs:string" minOccurs="0"/>
    </xs:sequence>
    <xs:attribute name="price" use="required">
      <xs:simpleType>
        <xs:restriction base="xs:float">
          <xs:minExclusive value="100"/>
        </xs:restriction>
      </xs:simpleType>
    </xs:attribute>
    <xs:attribute name="ISBN" type="xs:string" use="required"/>
    <xs:attribute name="year" use="optional">
      <xs:simpleType>
        <xs:restriction base="xs:gYear">
          <xs:minInclusive value="2000"/>
          <xs:maxInclusive value="2025"/>
        </xs:restriction>
      </xs:simpleType>
    </xs:attribute>
    <xs:attribute name="publicationdate" type="xs:date" use="optional"/>
  </xs:complexType>
```

```
        </xs:element>
      </xs:sequence>
    </xs:complexType>
  </xs:element>
</xs:schema>
```

- Validating with correct details which are under the restrictions is successfully validated without any errors
- Validating with invalid details of price under 100 and year outside 2000-2025 gives the following errors

- Cvc-maxInclusive-valid: Value '2050' Is Not Facet-valid With Respect To MaxInclusive '2025' For Type '#AnonType_yearbookbookstore'., Line '2', Column '77'.
- Cvc-attribute.3: The Value '2050' Of Attribute 'year' On Element 'book' Is Not Valid With Respect To Its Type, '#AnonType_yearbookbookstore'., Line '2', Column '77'.
- Cvc-minExclusive-valid: Value '67.774' Is Not Facet-valid With Respect To MinExclusive '1.0E2' For Type '#AnonType_pricebookbookstore'., Line '10', Column '49'.
- Cvc-attribute.3: The Value '67.774' Of Attribute 'price' On Element 'book' Is Not Valid With Respect To Its Type, '#AnonType_pricebookbookstore'., Line '10', Column '49'.

6.AIM: Write XSD and xml for the following structure.



This provides us with an understanding of the hierarchical structure of the XML

CODE: #XML code

```
<?xml version="1.0" encoding="UTF-8"?>
<EmployeeInfo>
  <Employee EmployeeNumber="101">
    <Name>SRP</Name>
    <Department>CSE</Department>
```

```
      <Telephone>9876543210</Telephone>
      <Email>SRP@gmail.com</Email>
    </Employee>
    <Employee EmployeeNumber="102">
      <Name>ISRPC</Name>
      <Department>EEE</Department>
      <Telephone>7654321098</Telephone>
      <Email>ISRPC@gmail.com</Email>
    </Employee>
  </EmployeeInfo>
```

#XSD code:

```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:element name="EmployeeInfo">
  <xs:complexType>
    <xs:sequence>
      <xs:element name="Employee" minOccurs="0" maxOccurs="unbounded">
        <xs:complexType>
          <xs:sequence>
            <xs:element name="Name" type="xs:string"/>
            <xs:element name="Department" type="xs:string"/>
            <xs:element name="Telephone" type="xs:string"/>
            <xs:element name="Email" type="xs:string"/>
          </xs:sequence>
          <xs:attribute name="EmployeeNumber" type="xs:integer" use="required"/>
        </xs:complexType>
      </xs:element>
```

```
        </xs:sequence>
      </xs:complexType>
    </xs:element>
    </xs:schema>
```

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